

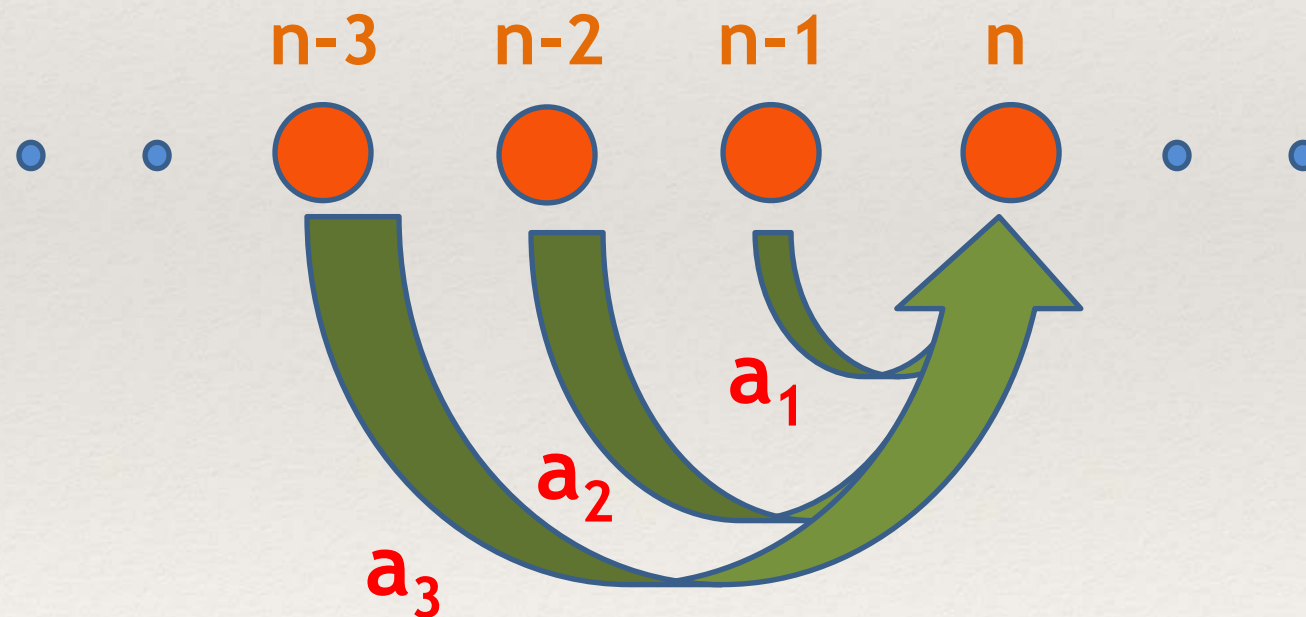
E9 205 Machine Learning for Signal Processing

Linear Predictive Analysis

22-08-2016

Linear Prediction

- ❖ Current sample expressed as a linear combination of past samples



Properties of LP

Error signal (for the optimal predictor) is orthogonal to the samples used in the predictor.

$$e[n] \perp \{x[n-1], \dots, x[n-N]\}$$

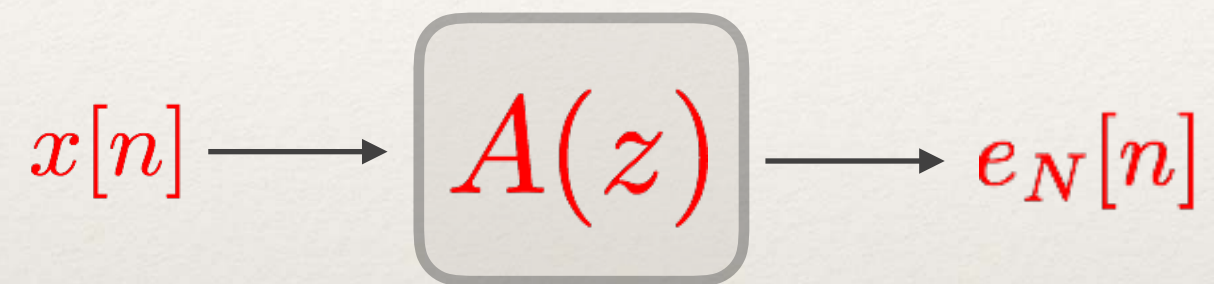
Using the orthogonality property $\rightarrow$ normal equations

$$\mathbf{R}\mathbf{a} = -\mathbf{r}$$

Autocorrelation matrix is Hermitian symmetric.

Properties of LP

Forward linear prediction filter



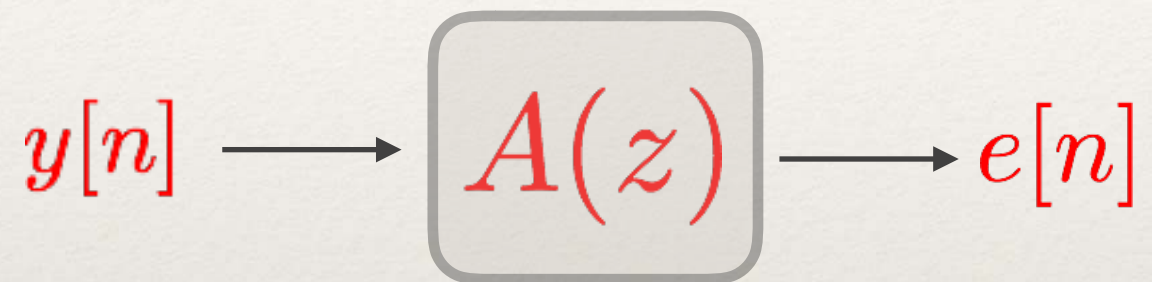
Properties of $A(z)$ - stability (all roots q)

$$|q| < 1$$

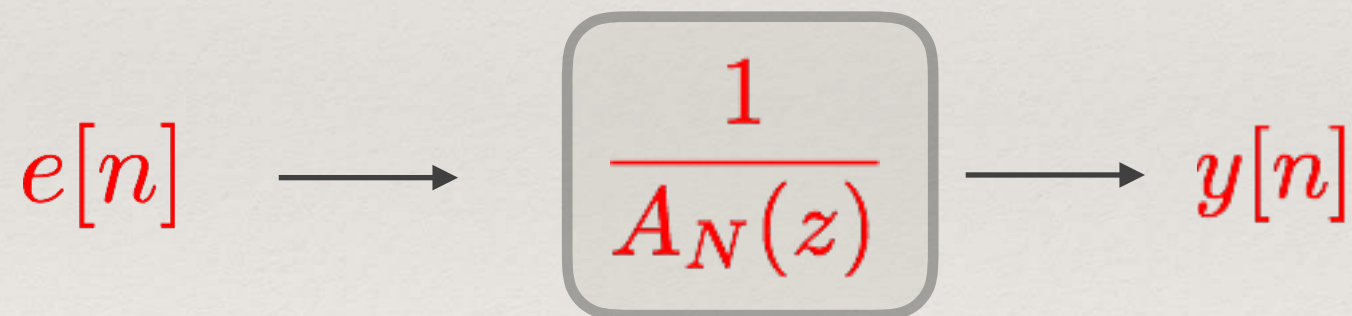
except for line spectral process $|R(k)| = R(0)$ *for some k*

Properties of LP

AR(N) process - Any WSS process which satisfies



Filter is stable - error signal is white

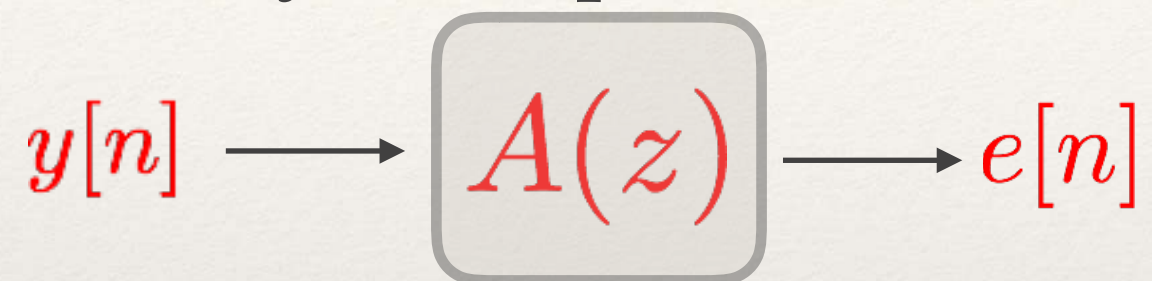


$$S_{yy}(f) = \frac{\epsilon_N}{|1 + \sum_{n=1}^N a_{N,n}^* e^{-j2\pi f n}|^2}$$

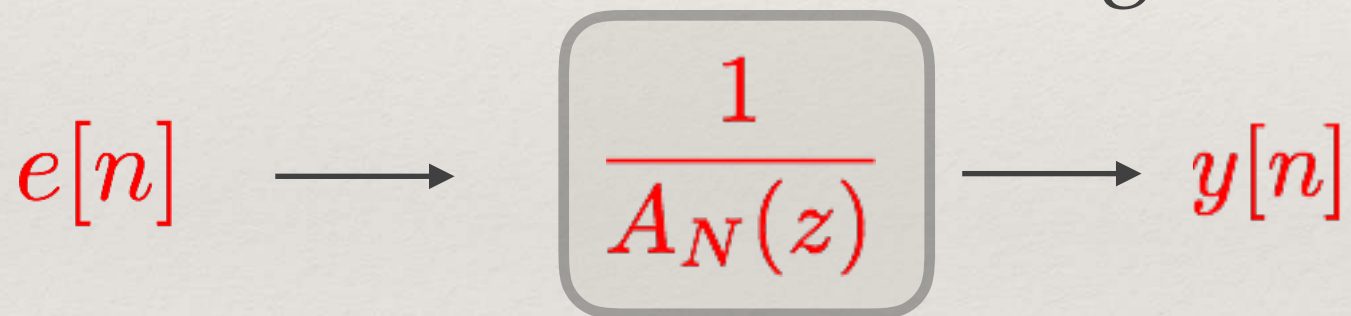
Approximating $x[n]$ by $y[n]$ i.e. $S_{xx}(f)$ with $S_{yy}(f)$

Properties of LP

AR(N) process - Any WSS process which satisfies



Filter is stable - error signal is white

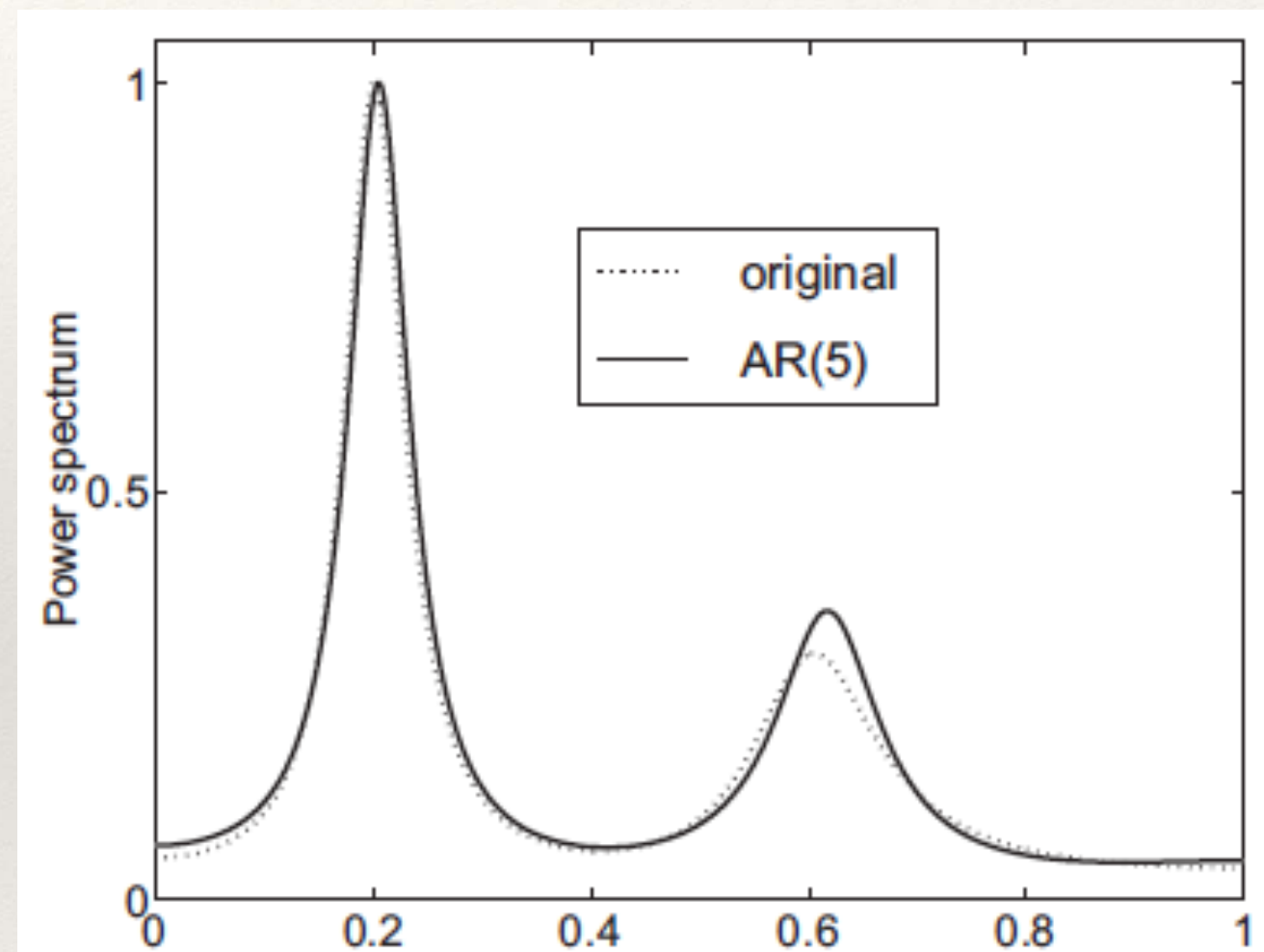
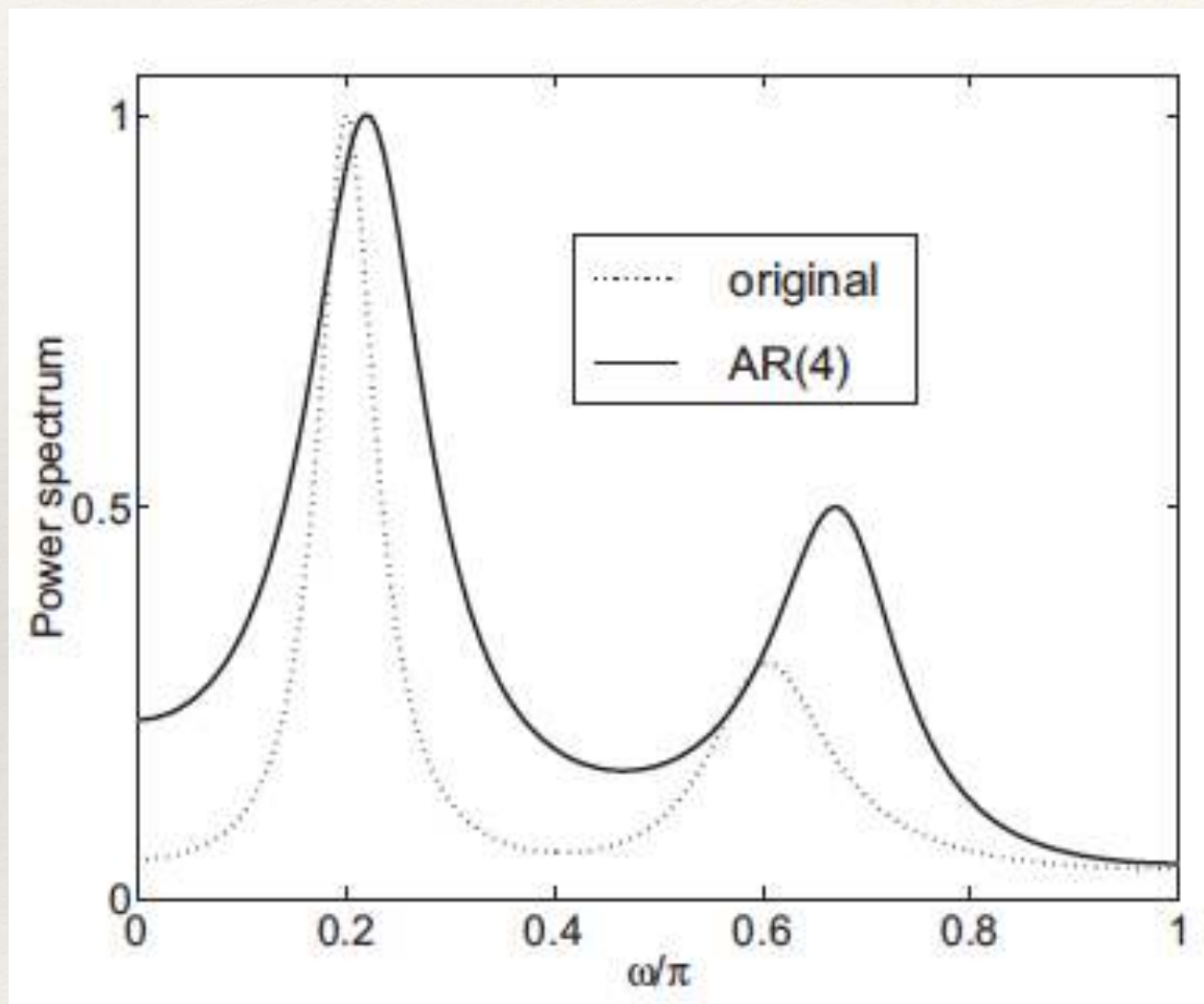


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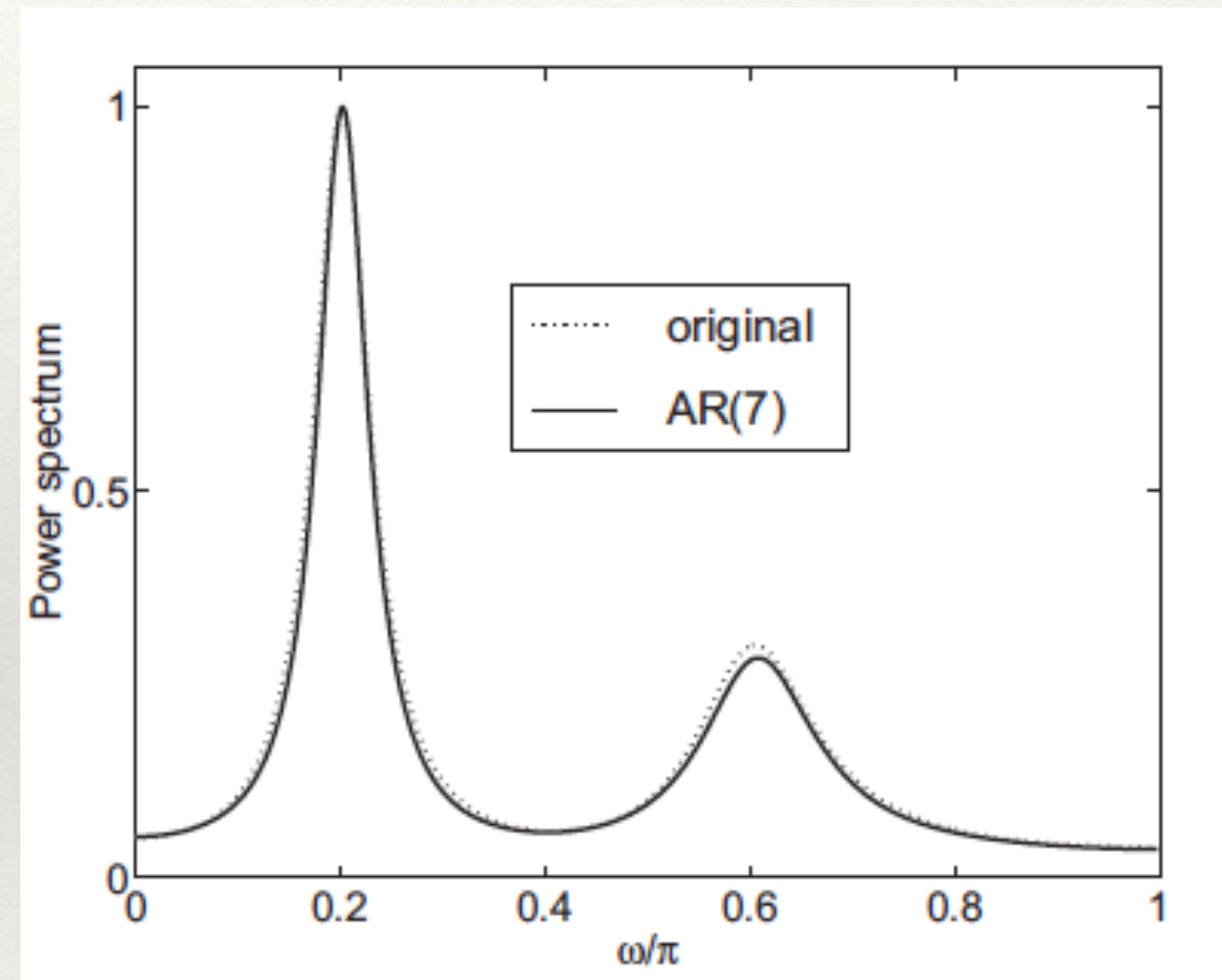
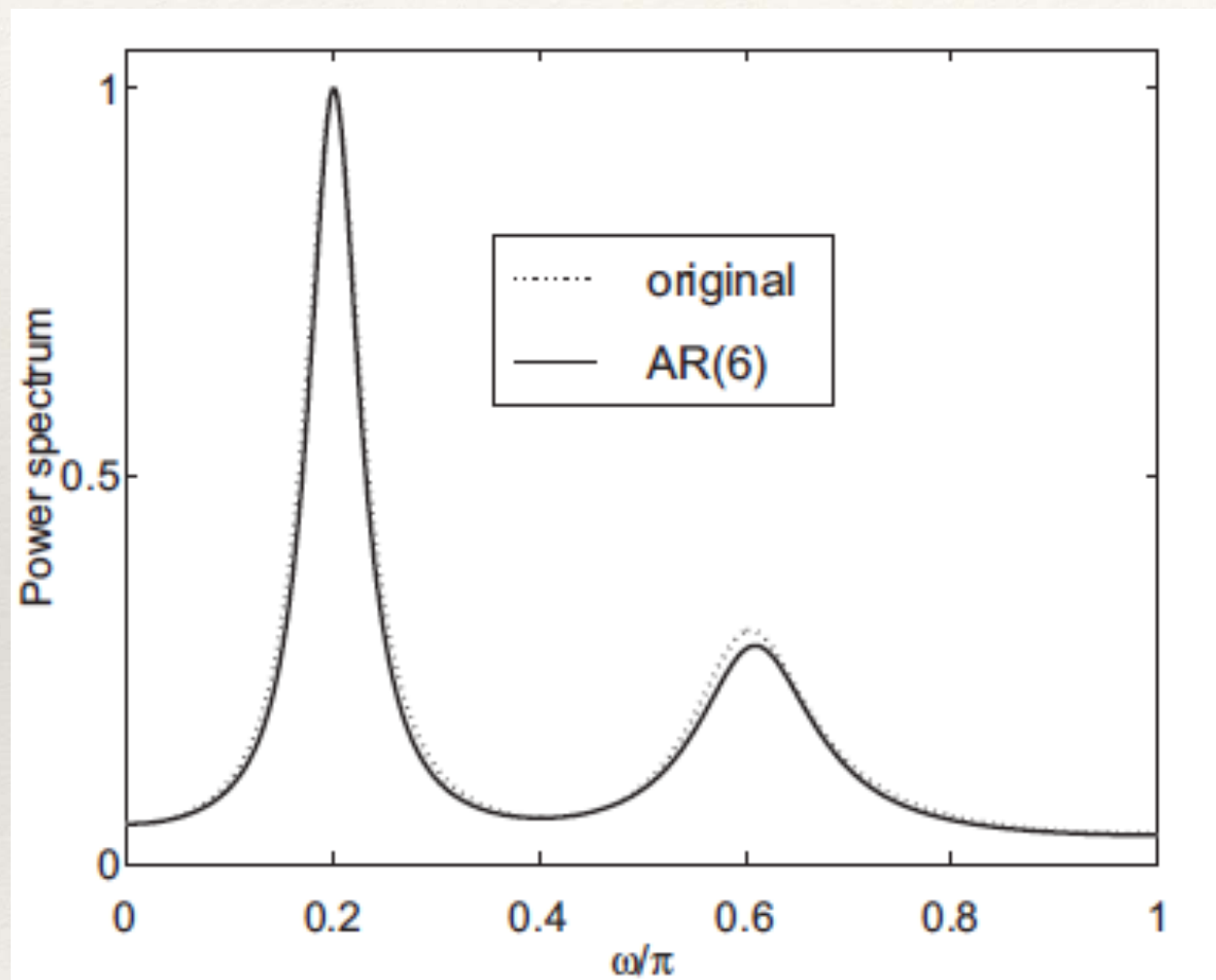
Approximating $x[n]$ by $y[n]$ i.e. $S_{xx}(f)$ with $S_{yy}(f)$

Autoregressive modeling

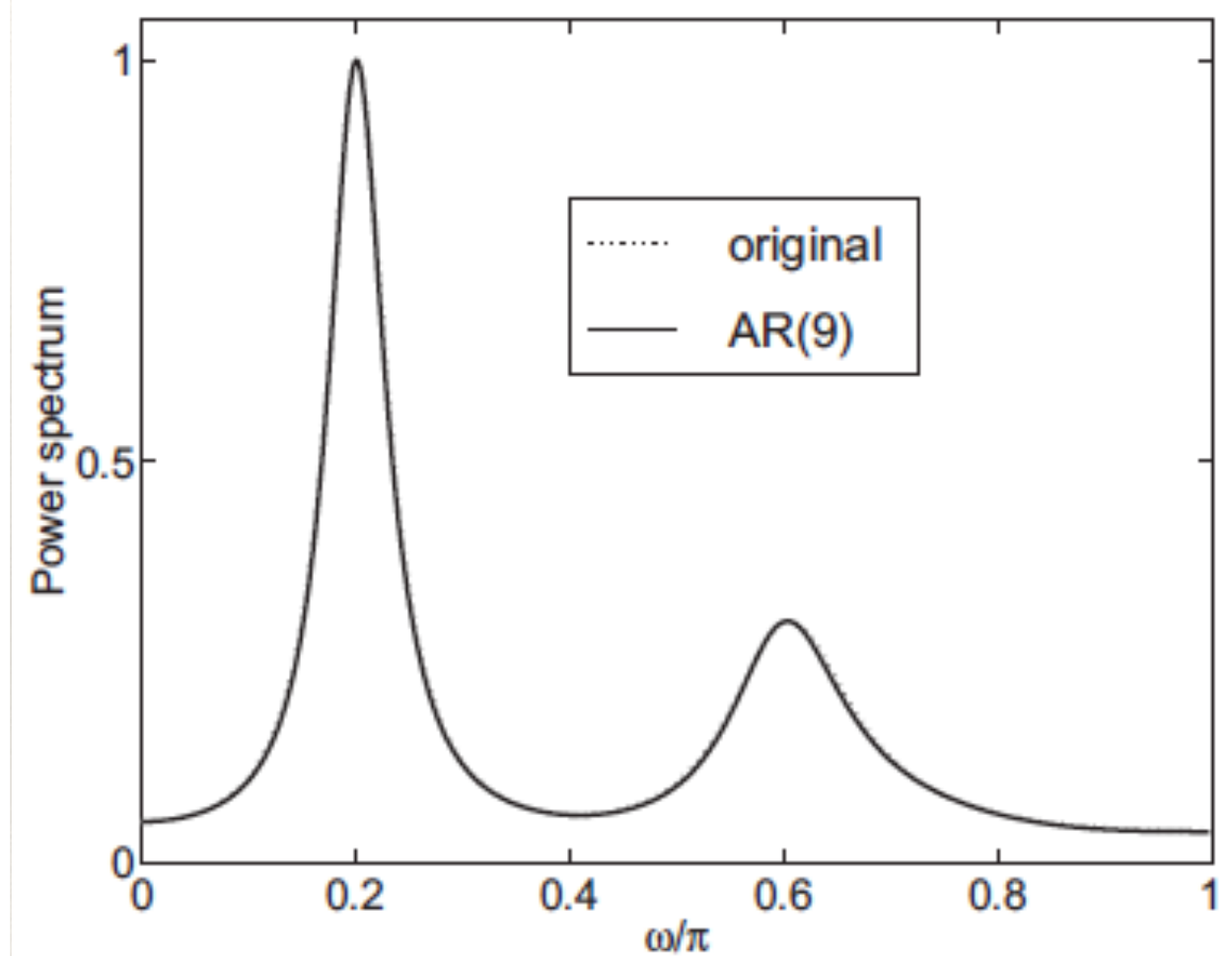
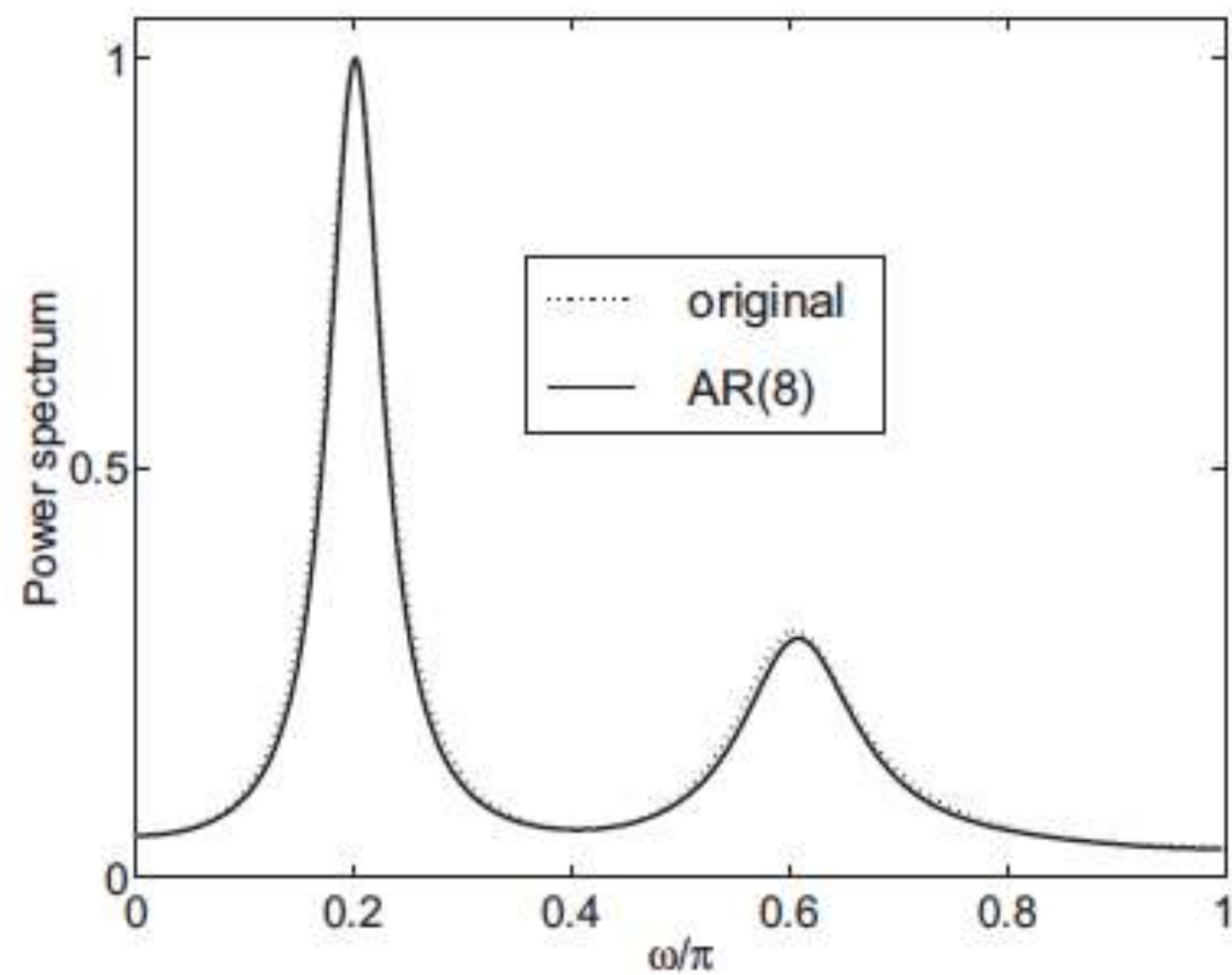
Properties of LP



Properties of LP

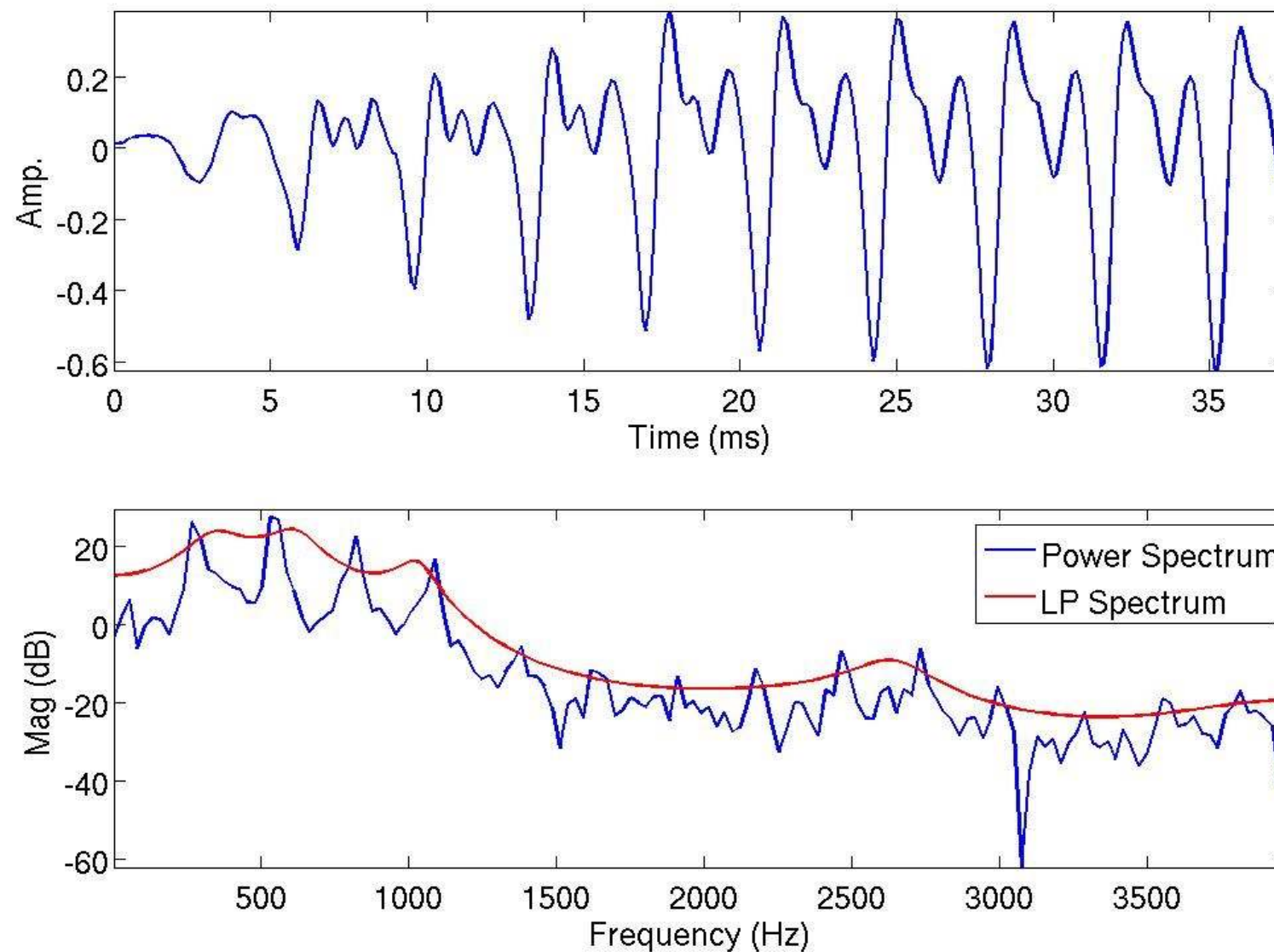


Properties of LP



Linear Prediction

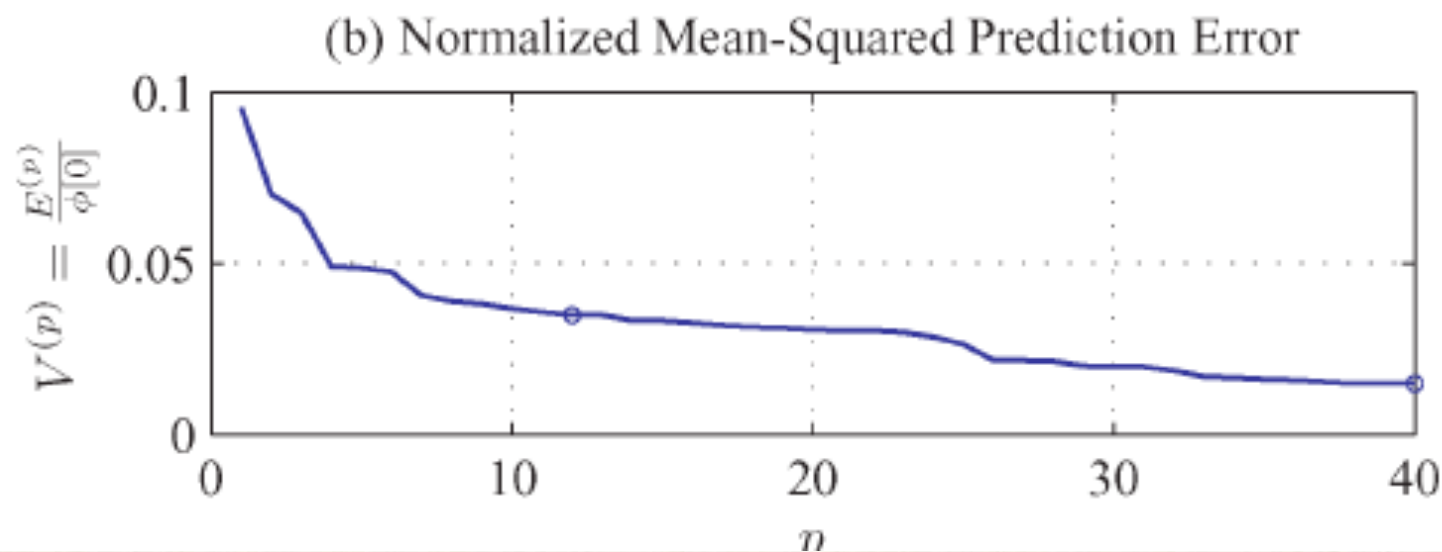
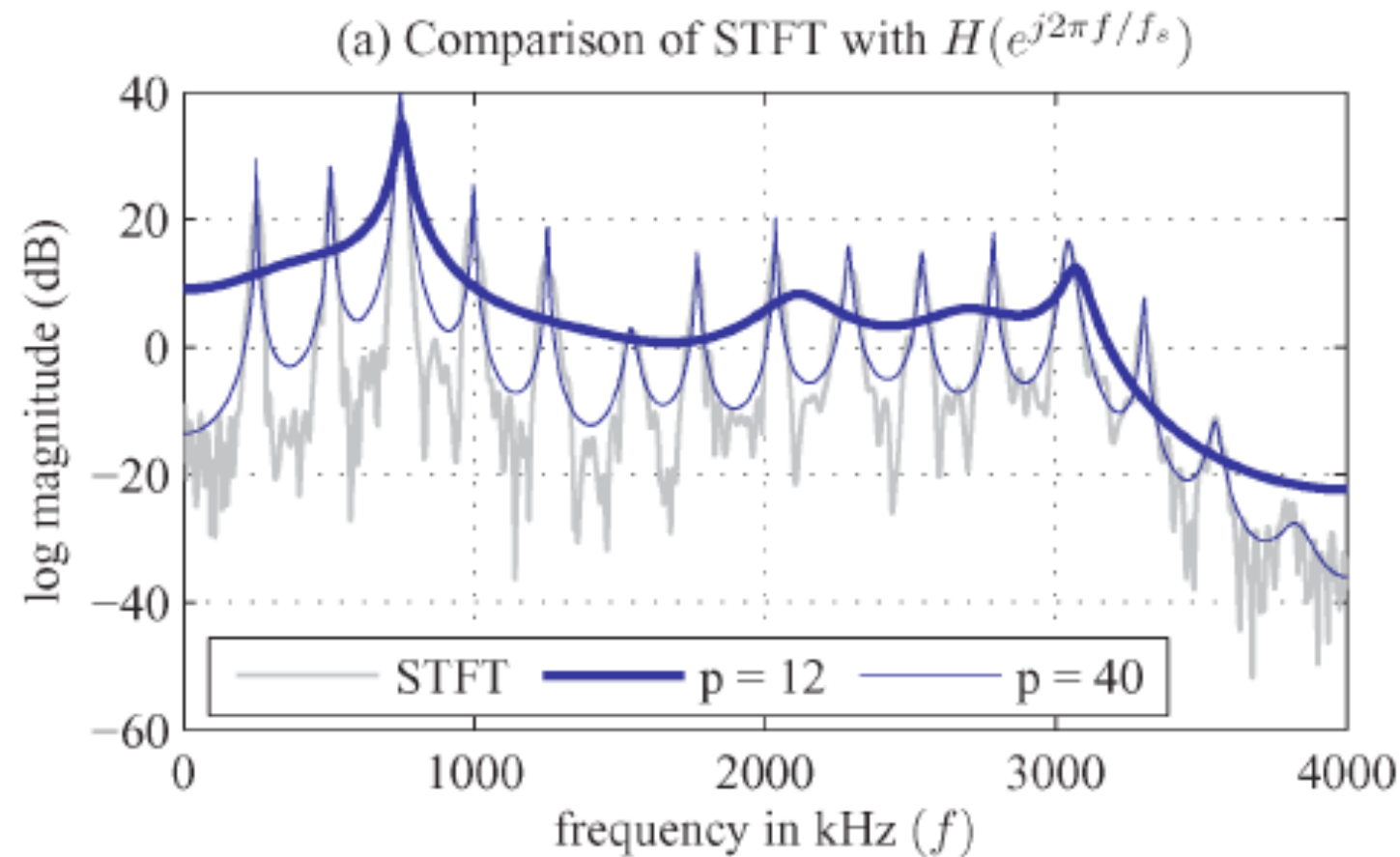
AR Model of the Power Spectrum of the Signal



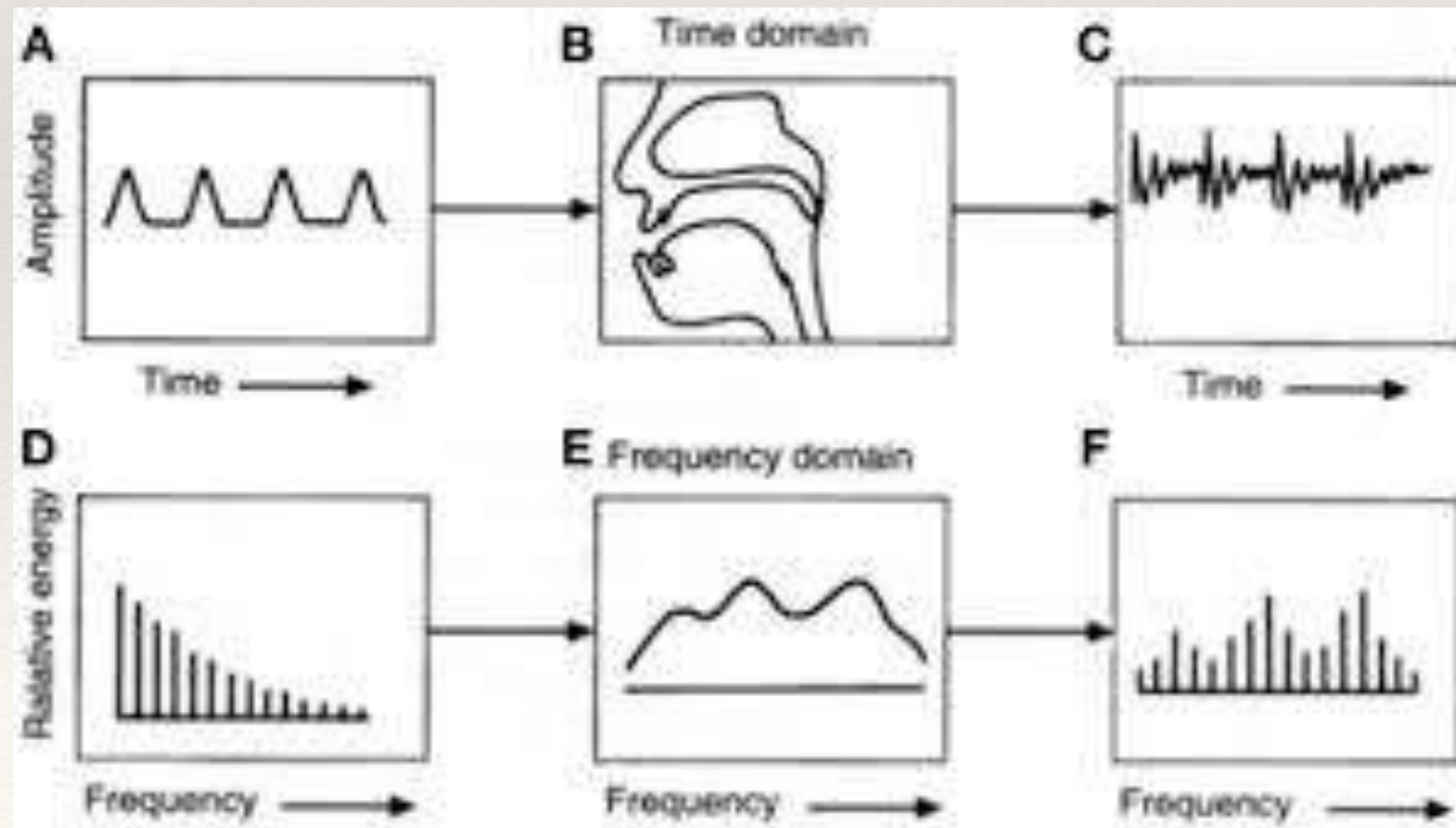
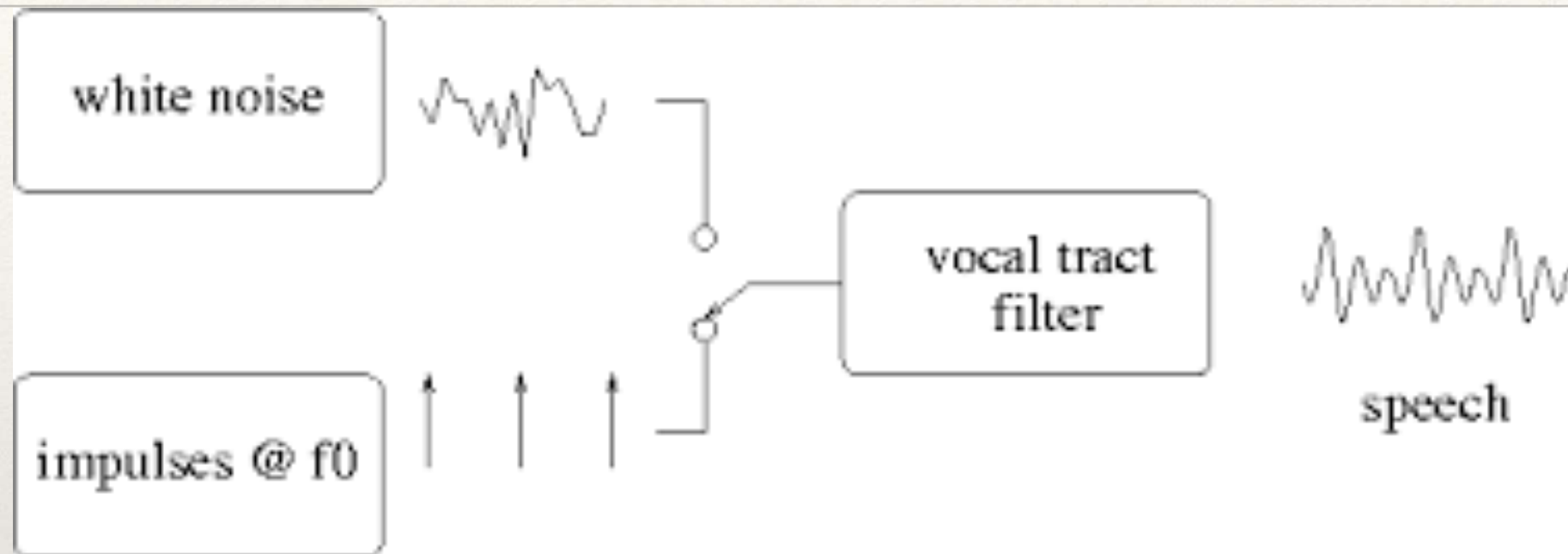
Applications of Autoregressive Modeling

- ❖ Economics - Macroeconomic variabilities
- ❖ Statistics - System Identification.
- ❖ Geophysics - Oil Exploration.
- ❖ Neurophysics - EEG signal analysis (rhythms)
- ❖ Speech Communication - Coding, Recognition.

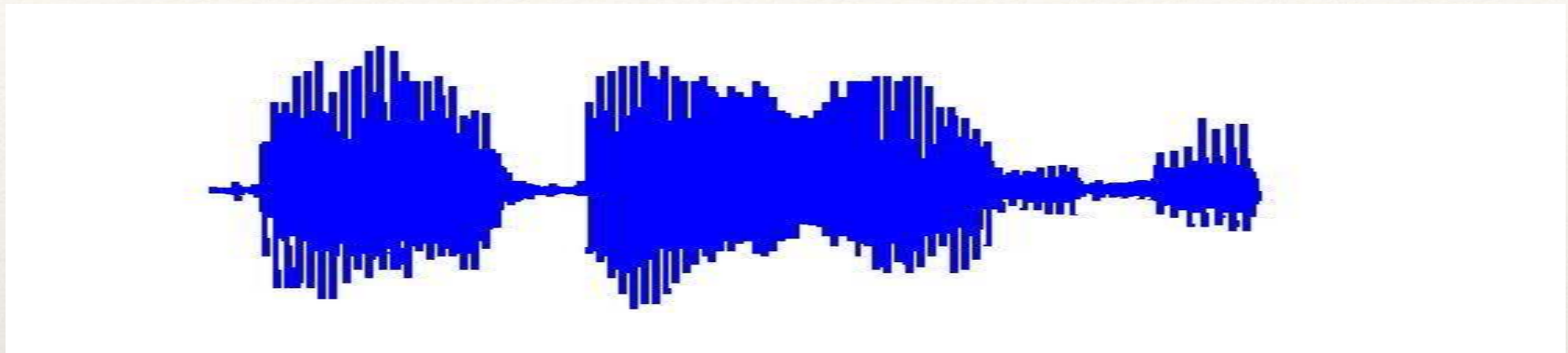
Linear Prediction for Speech



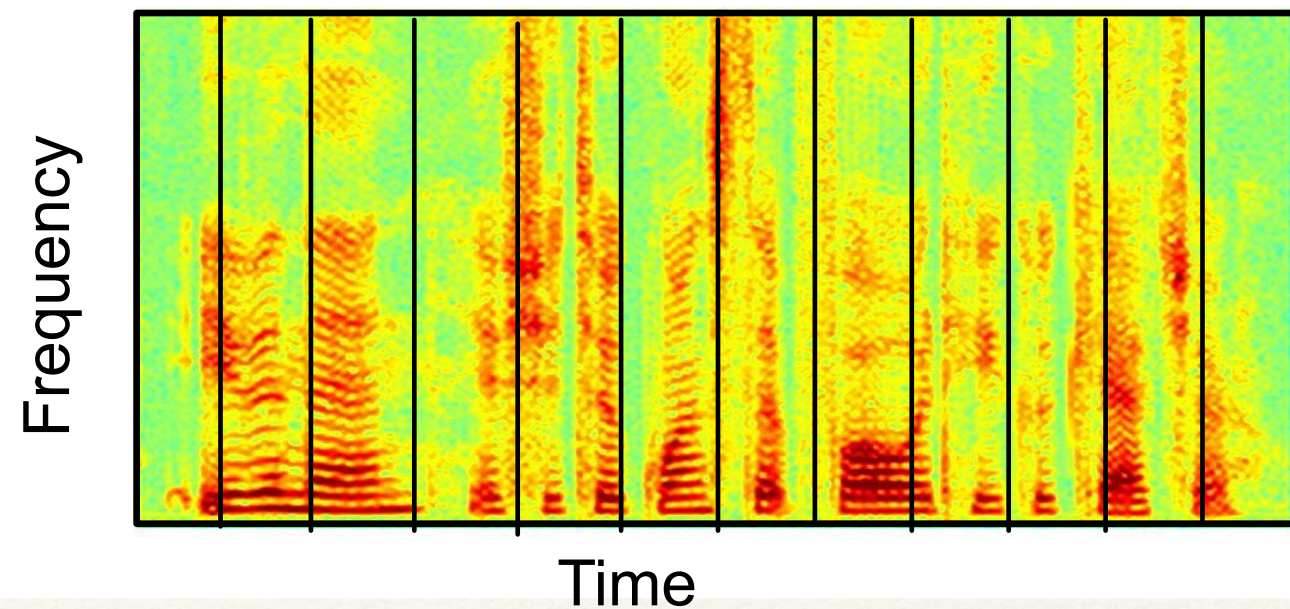
Source Filter Model of Speech



Feature Extraction for Speech/Audio

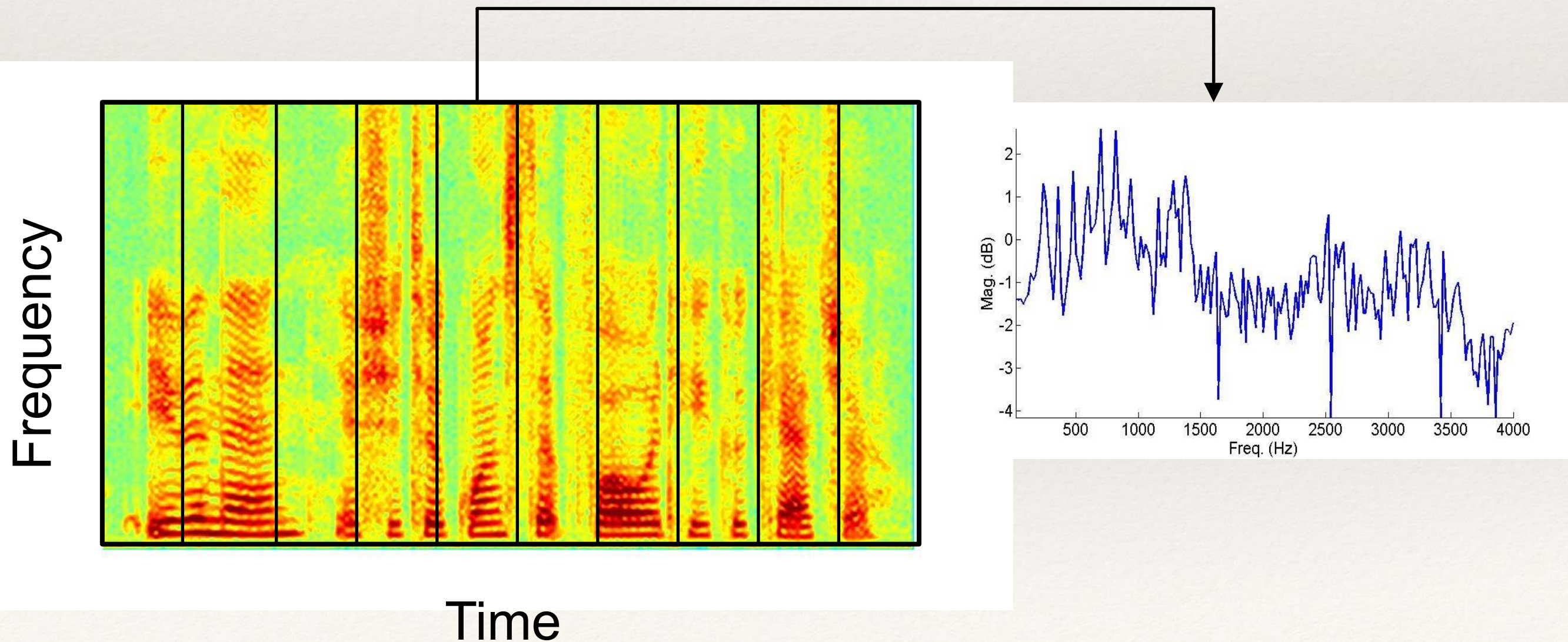


Conversion to Spectrogram



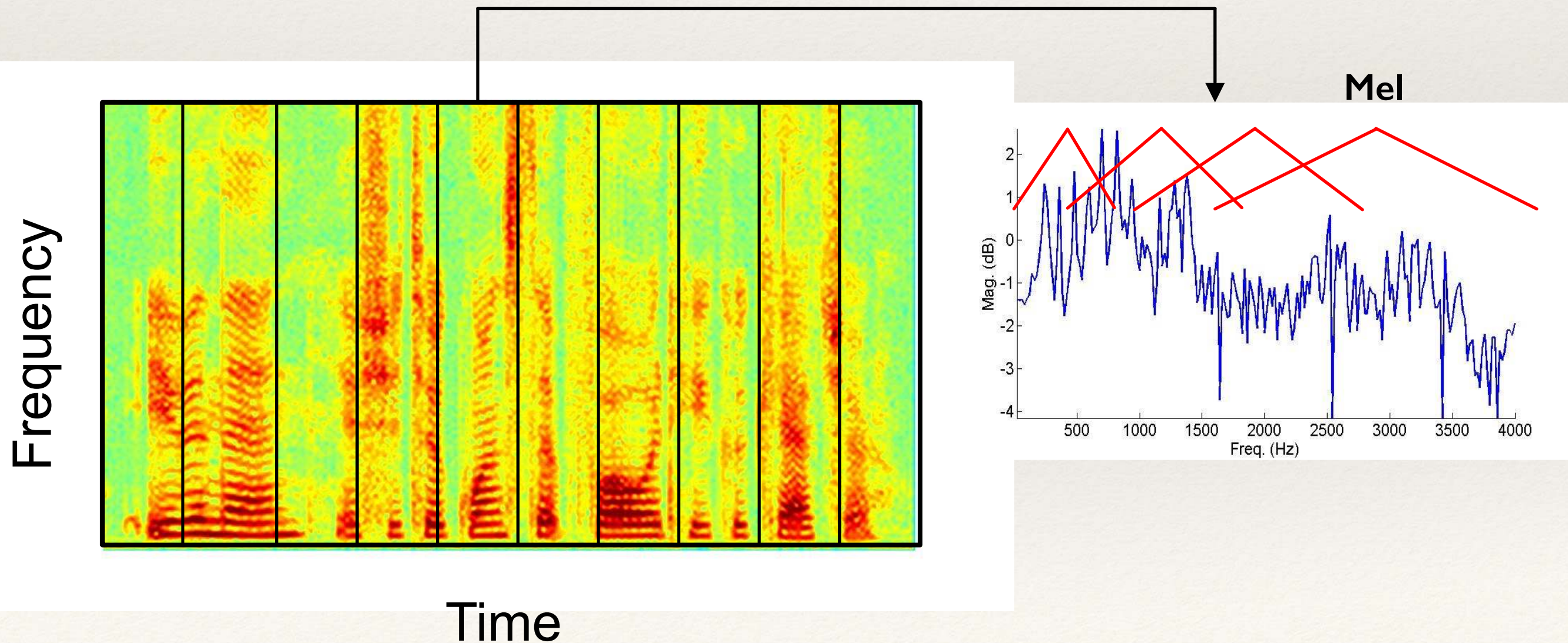
Feature Extraction for Speech/Audio

Integration to Mel-scale



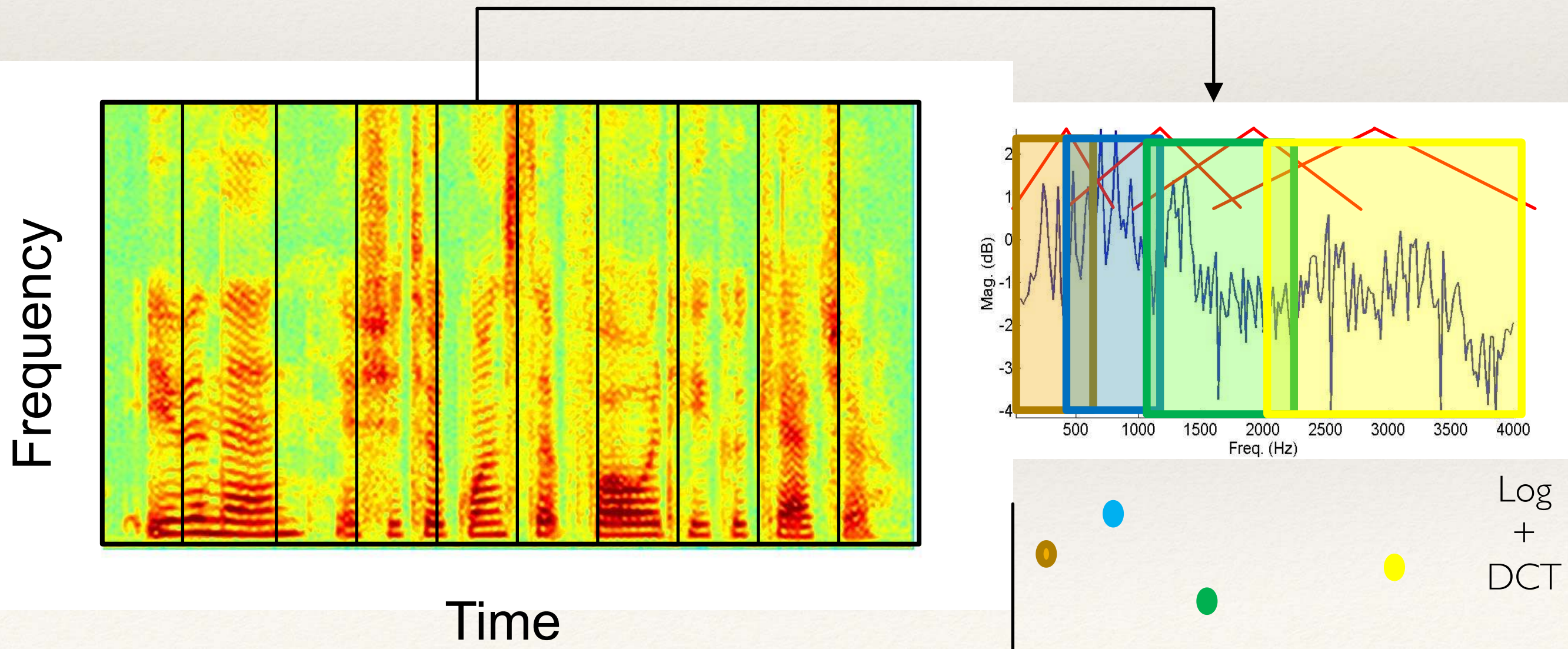
Feature Extraction for Speech/Audio

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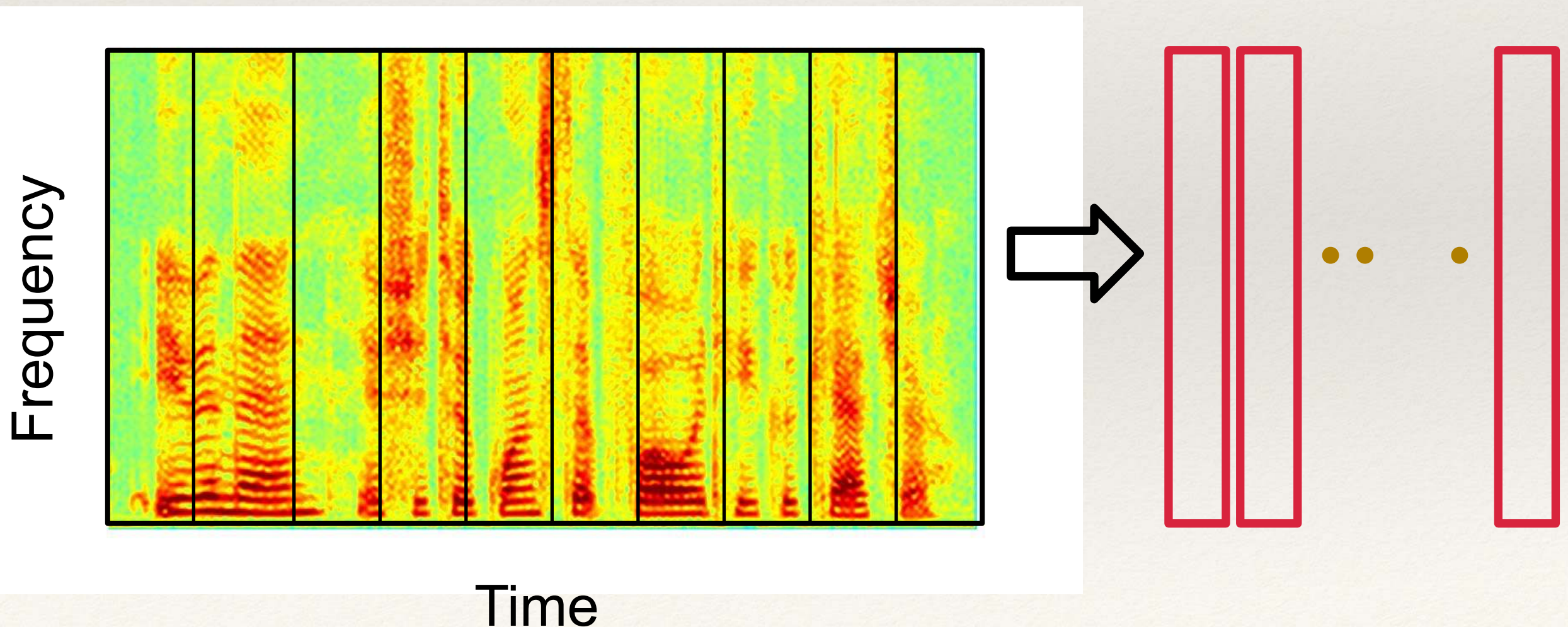
Feature Extraction for Speech/Audio

Integration to Mel-scale



Feature Extraction for Speech/Audio

Conversion to features - Mel frequency
cepstral coefficients (MFCC)

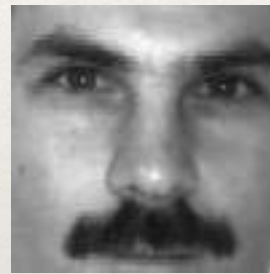


Recap so far ...

- ❖ Signal analysis - STFT
 - ❖ Choice of suitable window, time frequency resolution.
- ❖ STFT factorization
 - ❖ NMF - cost function, auxiliary function, divergence, applications in speech / audio.
- ❖ Signal Analysis - linear prediction
 - ❖ Orthogonality of error, normal equations, approximation with AR(N) process, autoregressive modeling.

Face Images (Assignment)

Normal Lighting Conditions



Occlusion

